

Sharp Local Fields as an Ultraviolet Scope Enlargement

Companion note: why a macroscopic Hamiltonian gap statement does not require $t \rightarrow 0$ fields

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Abstract

In constructive and axiomatic quantum field theory, the term “mass gap” ultimately refers to a spectral statement for the generator of time translations in the vacuum representation. A common presentation of the Yang–Mills mass-gap problem additionally demands the construction of sharp local operator-valued distributions (Wightman fields) and their ultraviolet localization properties. This note separates these requirements at the level of observable algebras. Throughout, “mass gap” is used in the sense of a Hamiltonian spectral gap above the vacuum.

We formalize two nested scopes. The first is a bounded macroscopic net $\mathcal{O}_\ell$ of local gauge-invariant observables at a fixed positive resolution scale $\ell > 0$. The second enlarges the observable class toward pointlike fields via a $\ell \downarrow 0$ limit (or equivalently $t \rightarrow 0$ in gradient-flow terminology). We explain why a Hamiltonian spectral gap can be proved and stated entirely within the macroscopic scope, and why the additional demand for sharp local fields should be regarded as an ultraviolet regularity upgrade rather than a primitive ingredient of the mass-gap statement on $\mathcal{O}_\ell$.

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1 Purpose and placement in the arc

Route 1 constructs, at a fixed macroscopic scale $\ell > 0$, an Osterwalder–Schrader (OS) positive Euclidean theory on a bounded local net of gauge-invariant observables and proves exponential clustering and a Hamiltonian spectral gap in the OS reconstruction. Paper 4 isolates the ultraviolet (UV) identification inputs needed to interpret this as a continuum statement at fixed $\ell > 0$.

The remaining source of predictable confusion is the relationship between:

- (i) a spectral gap statement for the reconstructed Hamiltonian on the vacuum representation generated by $\mathcal{O}_\ell$, and
- (ii) the construction of sharp local operator-valued distributions obtained by an ultraviolet limit $\ell \downarrow 0$ (or $t \rightarrow 0$).

This note clarifies that (ii) is a strictly stronger ultraviolet regularity requirement than what is needed for (i). The goal is not to deny the importance of sharp fields in the classical Wightman framework, but to make the scope distinction explicit and technically neutral.

Remark 1.1 (Scope discipline). All statements in this note are formulated in standard mathematical-physics language. No model-dependent estimates are proved here. The note records structural relationships between different levels of observable algebras and the spectral-gap property in the vacuum representation.

What is made explicit in this note

The intent is to make two points checkable and easy to cite when reading Route 1:

- (S1) a Hamiltonian spectral gap is a statement about a specific vacuum representation generated by a specified observable algebra (here the macroscopic net $\mathcal{O}_\ell$ at fixed $\ell > 0$);
- (S2) the passage from a bounded net $\mathcal{O}_\ell$ to pointlike fields involves additional ultraviolet regularity inputs (energy bounds, phase-space/nuclearity, scaling limits) studied in algebraic QFT, and is not forced by the existence of a macroscopic spectral gap.

2 Two levels of locality: nets of bounded observables versus point-like fields

2.1 Bounded local nets at fixed resolution

Definition 2.1 (Bounded local net at scale ℓ). Fix $\ell > 0$. A *bounded local net at scale ℓ* is an assignment

$$\Lambda \mapsto \mathcal{O}_\ell(\Lambda)$$

from bounded regions $\Lambda \subset \mathbb{R}^4$ to unital $*$ -algebras of bounded observables, satisfying isotony and translation covariance (as in Papers 2–4). Let

$$\mathcal{A}_\ell := \bigcup_{\Lambda} \mathcal{O}_\ell(\Lambda)$$

be the global $*$ -algebra.

In Yang–Mills theory, the physically standard choice of observables is gauge invariant (Wilson loops, bounded functions of plaquette variables, or bounded functionals of smeared field-strength composites). The Route 1 construction is formulated on such a bounded, gauge-invariant macroscopic net.

Table 1: Two nested scopes: macroscopic bounded nets at fixed resolution versus ultraviolet sharp-field upgrades.

	Macroscopic scope (fixed $\ell > 0$)	Sharp-field scope ($\ell \downarrow 0$)
Observable class	bounded local observables in $\mathcal{O}_\ell(\Lambda)$	(typically unbounded) operator-valued distributions $\phi(f)$
Localization	region localization at resolution ℓ	pointlike/test-function localization
Core axioms used	OS positivity on bounded algebra; time-clustering for bounded observables	Wightman locality/temperedness; ultraviolet domain/energy control
Typical inputs	contraction/cluster bounds at scale ℓ	energy bounds, nuclearity/phase space, scaling limits
Primary output	Hamiltonian spectral gap on the vacuum sector generated by $\mathcal{O}_\ell$	existence/regularity of pointlike fields and ultraviolet locality properties

2.2 Pointlike fields as an ultraviolet extension

In the Wightman framework, “local fields” are operator-valued distributions ϕ such that $\phi(f)$ is an (unbounded) operator for each test function f with compact support, and locality is expressed by commutativity at spacelike separation. The existence of such fields requires substantially more ultraviolet control than the existence of a bounded net.

Definition 2.2 (Ultraviolet scope enlargement). Let $\{\mathcal{O}_\ell\}_{\ell>0}$ be a family of bounded local nets with natural inclusion maps as ℓ decreases (interpreting smaller ℓ as higher resolution). An *ultraviolet (UV) extension to sharp fields* is an additional structure providing operator-valued distributions ϕ such that, for suitable test functions f , the smeared field $\phi(f)$ is recovered as a limit (in an appropriate topology) of observables built from the nets $\mathcal{O}_\ell$ as $\ell \downarrow 0$.

Remark 2.3 (Why this is a scope enlargement). At fixed $\ell > 0$, the net $\mathcal{O}_\ell$ only tests the theory at a bounded resolution. Passing to $\ell \downarrow 0$ changes the class of admissible probes and introduces ultraviolet regularity questions (energy bounds, short-distance scaling, existence of distributional limits) that are not forced by the spectral gap property at a fixed ℓ .

3 Mass gap as a spectral property in the vacuum representation

3.1 Vacuum representation and Hamiltonian

In the algebraic formulation, a vacuum state ω on a quasilocal algebra determines a GNS triple $(\pi, \mathcal{H}, \Omega)$ and a unitary representation of translations $U(a)$ implementing the automorphisms τ_a . A *Hamiltonian* is the generator of the time-translation subgroup $t \mapsto U(t, \mathbf{0})$.

Definition 3.1 (Hamiltonian spectral gap on a bounded net). Let $(\pi, \mathcal{H}, \Omega)$ be the vacuum representation of a translation-covariant net with positive energy. Let $H \geq 0$ be the generator of time translations. We say the theory has a *Hamiltonian spectral gap* $m > 0$ above the vacuum if

$$\text{Spec}(H) \subset \{0\} \cup [m, \infty),$$

and $\ker H = \mathbb{C}\Omega$.

Remark 3.2 (Terminology). In this note, “mass gap” refers to the Hamiltonian spectral-gap property in Definition 3.1 for the vacuum representation generated by the specified observable net. No claim is made here about additional relativistic spectral structure (for example, the full joint spectrum of energy–momentum) beyond what is encoded by the time-translation generator in the given representation.

This definition makes no reference to pointlike fields. It is a statement about the spectrum of the representation of time translations in the vacuum sector generated by the chosen observable algebra.

3.2 Spectral gap is determined by correlation decay of bounded observables

Paper 3 records the standard OS semigroup mechanism: if OS positivity is available and Euclidean time-separated correlations of bounded observables decay exponentially, then H has a spectral gap. The extraction depends only on the observable algebra used to build the OS Hilbert space.

Remark 3.3 (Gauge-invariant observables). In gauge theories, the physically standard local algebra is gauge invariant. A mass gap formulated in terms of H in the vacuum representation of the gauge-invariant net is therefore the natural spectral formulation.

4 Why sharp local fields are an ultraviolet regularity upgrade

4.1 What additional structure sharp fields require

Constructing pointlike fields (or proving the existence of operator-valued distributions affiliated with local algebras) typically requires ultraviolet estimates beyond those needed for a macroscopic spectral gap, for example:

- (U1) **Uniform energy bounds:** control of norms like $\|(1 + H)^k \pi(A_\ell)\Omega\|$ uniformly as $\ell \downarrow 0$ for suitable families $A_\ell \in \mathcal{O}_\ell$.
- (U2) **Short-distance scaling bounds:** estimates ensuring that smeared observables have limits as the support shrinks.
- (U3) **Phase-space/nuclearity conditions:** compactness or nuclearity properties that allow reconstruction of pointlike fields from the net.

These are precisely the types of assumptions that are invisible at a fixed positive resolution scale ℓ .

Remark 4.1 (Pointlike fields from Haag–Kastler nets require extra hypotheses). In algebraic QFT, reconstructing pointlike fields from a net of local algebras is a separate problem that typically requires phase-space or energy-bound hypotheses. For example, Fredenhagen–Hertel recover pointlike localized fields from local algebras under additional regularity assumptions, and related constructions for conformal nets and short-distance analysis appear in work of Fredenhagen–Jörß and in the scaling-algebra approach of Bostelmann–D’Antoni–Morsella. Nuclearity/phase-space conditions controlling the local density of states (in the sense of Buchholz–Wichmann) provide another standard route to ultraviolet control. See [2, 3, 4, 5].

4.2 A clean separation principle

The following elementary proposition formalizes the sense in which a spectral gap statement is stable under *restriction of scope*.

Proposition 4.2 (Restriction stability of the spectral gap). *Let $H \geq 0$ be self-adjoint on $\mathcal{H}$ with $\text{Spec}(H) \subset \{0\} \cup [m, \infty)$ and $\ker H = \mathbb{C}\Omega$. Let $\mathcal{H}_0 \subset \mathcal{H}$ be a closed subspace invariant under e^{-tH} (equivalently under $U(t, \mathbf{0})$). Then the restricted generator $H|_{\mathcal{H}_0}$ has the same gap:*

$$\text{Spec}(H|_{\mathcal{H}_0}) \subset \{0\} \cup [m, \infty), \quad \ker(H|_{\mathcal{H}_0}) = \mathbb{C}\Omega.$$

Proof. Let P_0 denote the projection onto $\ker H = \mathbb{C}\Omega$. The spectral inclusion is immediate because $\text{Spec}(H|_{\mathcal{H}_0}) \subset \text{Spec}(H)$ for an invariant reducing subspace. Vacuum uniqueness follows since $\Omega \in \mathcal{H}_0$ and $\mathcal{H}_0 \cap \ker H = \mathbb{C}\Omega$. $\square$

Proposition 4.3 (Extensions and the macroscopic vacuum sector). *Let $\mathcal{A}_0 \subset \mathcal{A}$ be unital $*$ -algebras and let ω be a state on $\mathcal{A}$. Let $(\pi, \mathcal{H}, \Omega)$ be the GNS triple of ω and set*

$$\mathcal{H}_0 := \overline{\pi(\mathcal{A}_0)\Omega} \subset \mathcal{H}.$$

Then $(\pi|_{\mathcal{A}_0}, \mathcal{H}_0, \Omega)$ is (canonically) the GNS triple of the restricted state $\omega|_{\mathcal{A}_0}$.

Moreover, if a time-translation group $U(t) = e^{itH}$ implements automorphisms of $\mathcal{A}$ and leaves $\mathcal{A}_0$ invariant, then $\mathcal{H}_0$ is $U(t)$ -invariant and the spectral properties of H on $\mathcal{H}_0$ are determined by the correlations of $\omega|_{\mathcal{A}_0}$. In particular, a spectral-gap statement proved in the vacuum representation generated by $\mathcal{A}_0$ is unaffected by enlarging the observable algebra, except for the possibility of additional orthogonal sectors not generated from $\mathcal{A}_0$.

Proof. The first statement is standard: the cyclic subspace generated by $\pi(\mathcal{A}_0)\Omega$ realizes the GNS construction for $\omega|_{\mathcal{A}_0}$ by restriction. If $\mathcal{A}_0$ is invariant under the implemented automorphisms, then $U(t)\pi(A)\Omega = \pi(\tau_t A)\Omega \in \pi(\mathcal{A}_0)\Omega$ for $A \in \mathcal{A}_0$, hence $\mathcal{H}_0$ is invariant. The final sentence is an interpretation: the vacuum-sector spectral information accessible to $\mathcal{A}_0$ depends only on the restriction $\omega|_{\mathcal{A}_0}$ and its GNS representation. $\square$

Remark 4.4 (Scope-contained gap statements and ultraviolet extensions). Proposition 4.2 states that once a Hamiltonian gap is proved in a given vacuum representation, restricting to a smaller invariant subspace (generated by a smaller observable algebra) cannot destroy that gap. Proposition 4.3 records the complementary fact: if a state ω is defined on a larger algebra $\mathcal{A}$ and one restricts to a subalgebra $\mathcal{A}_0$, then the GNS representation of $\omega|_{\mathcal{A}_0}$ is realized canonically as the cyclic subspace $\mathcal{H}_0 = \overline{\pi(\mathcal{A}_0)\Omega}$ inside the GNS space of ω .

Accordingly, when Route 1 asserts a Hamiltonian spectral gap on the macroscopic net $\mathcal{O}_\ell$, it is a statement about the vacuum representation generated by $\mathcal{O}_\ell$ (Definition 3.1). Any ultraviolet extension that agrees on $\mathcal{O}_\ell$ contains the same macroscopic vacuum sector as a subrepresentation, and the gap on that sector is unchanged. This does *not* assert that every ultraviolet extension preserves the gap as a global property: an extension may introduce additional orthogonal sectors not generated from $\mathcal{O}_\ell$, and ruling these out requires additional ultraviolet control.

5 Interpreting the $t \rightarrow 0$ demand as a scope change

5.1 Clay-style formulations and Wightman fields

Many standard presentations of the Yang–Mills mass-gap problem are phrased in Wightman terms: construct a relativistic QFT with operator-valued distributions satisfying locality, covariance, and positivity, and show a mass gap. Those formulations implicitly package together:

- (C1) existence of a local net / vacuum representation with positive energy,
- (C2) a spectral gap above the vacuum,
- (C3) ultraviolet localization sufficient to support pointlike fields (tempered distributions).

Route 1 addresses (C1)–(C2) on a fixed macroscopic net $\mathcal{O}_\ell$. The passage to (C3) requires an ultraviolet limit $\ell \downarrow 0$ and additional bounds of the type described in the previous section.

5.2 Why this does not undermine the macroscopic gap statement

The macroscopic gap statement is a property of the vacuum representation generated by $\mathcal{O}_\ell$. It is extracted from OS positivity and exponential clustering of the corresponding Euclidean correlation functions. Whether the theory admits an ultraviolet extension to pointlike fields is a separate structural question about enlarging the observable class.

Equivalently: the spectral gap is determined by the semigroup e^{-tH} acting on the Hilbert space built from $\mathcal{O}_\ell$; it is not logically dependent on the existence of distribution-valued field coordinatizations.

6 What would be required to upgrade to sharp fields

This section records, neutrally, what an ultraviolet upgrade would entail if one wants a Wightman-type completion.

- (E1) **A shrinking-resolution family.** A coherent family of nets $\mathcal{O}_\ell$ for $\ell \downarrow 0$ with compatibility maps.
- (E2) **Uniform control.** Estimates uniform in ℓ that prevent norm blow-up of candidate field approximants.
- (E3) **Distributional limits.** Existence of limits in a topology compatible with test functions and locality.
- (E4) **Temperedness / covariance.** Growth bounds ensuring the resulting distributions are tempered and transform correctly.

These requirements are standard in spirit but technically demanding in four-dimensional gauge theory. The Route 1 argument intentionally isolates the mass-gap mechanism at fixed ℓ from these ultraviolet completion questions.

7 Conclusion

A Hamiltonian spectral gap is a statement about the vacuum representation of an observable algebra. It can be constructed and verified on a bounded macroscopic net $\mathcal{O}_\ell$ at fixed positive resolution $\ell > 0$, using OS positivity and Euclidean exponential clustering. The further demand for sharp local operator-valued distributions is an ultraviolet scope enlargement requiring additional regularity inputs that are logically independent of the spectral-gap extraction on $\mathcal{O}_\ell$.

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